

PERSISTENT FINITE DETECTION OF OFF-CRITICAL ZEROS IN THE WEIL–CCM HIERARCHY

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ABSTRACT. Recent finite Guinand–Weil dictionary results identify real-even centered Galerkin values with exact sums over the nontrivial zeros of the Riemann zeta function. We address the converse detection problem: if a hypothetical nontrivial zero lies off the critical line, must that failure become visible in an actual canonical finite centered matrix? We prove that it does. First, on a fixed common support envelope, the complete zeta Weil form satisfies an asymmetric quantitative continuity estimate controlled by a single inverse-square zero mass. Second, every compactly supported C^2 test supported strictly inside a positive aperture admits a finite centered Fourier approximation which can be corrected in exactly the three modes $-1, 0, 1$ so that the zeroth, first, and second centered coefficient moments vanish, while preserving the two L^1 error quantities consumed by the Weil estimate. Consequently every hypothetical off-critical zero forces a nonzero vector in a canonical finite boundary-flat sector with strictly negative quadratic value. The three moment constraints have exact complex codimension three, so the first possible nonzero obstruction occurs at $N \geq 2$.

We then prove exact centered nesting. Central zero extension preserves every centered moment, the represented hard-window finite function, and the canonical quadratic value. Hence one finite negative witness at aperture L persists, with unchanged quadratic value, in every larger centered truncation at the same aperture. As a corollary, universal nonnegativity of the canonical quadratic forms restricted to all boundary-flat finite sectors is equivalent to the critical-line statement. These are counterexample-detection and structural equivalence theorems; they do not establish the required nonnegativity and do not prove the Riemann hypothesis. The detection-and-persistence core is formally verified in Lean 4; the final RH-equivalent criterion is stated as a paper-level corollary of that core.

1. INTRODUCTION

Weil’s explicit formula admits a positivity formulation equivalent to the Riemann hypothesis, and the associated quadratic functional has been studied from variational, finite-dimensional, Hermitian-form, and operator-theoretic viewpoints for decades [13, 14, 1, 2, 12]. Recent work has sharpened the finite side of this picture. Connes and van Suijlekom analyze structured finite matrices attached to real-even quadratic forms [5]; Connes, Consani, and Moscovici restrict the localized Weil form to centered low-frequency spaces in the zeta spectral-triple program [4]; and Groskin

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gives an exact finite Guinand–Weil dictionary in which real-even Galerkin values are represented by complete sums over the nontrivial zeta zeros [6].

Two companion constructions isolate different aspects of possible off-critical behavior. A pair-local Gram construction gives an explicit finite negative certificate for one visible off-critical reflection pair, but deliberately leaves open the masking problem created by the remaining zero population [7]. A complete two-translate construction avoids that masking issue by building the full zero population into a 2×2 Weil matrix and proves that every hypothetical off-critical zero forces a strict negative complete direction [8]. Separately, the exact finite dictionary was reduced to one scalar nonsmooth defect and then closed; that manuscript ended with the following quantitative question: must a negative complete detector be approximable strongly enough by a centered finite dictionary to force finite negativity [9]?

The present paper closes the *existence* part of that question. It does not give an effective numerical upper bound for the required finite size. Instead, it proves the qualitative implication

- (1) off-critical zero $\implies$ negative canonical finite Weil–CCM direction.

The second result is a rigidity statement about finite size. Once such a direction exists at one centered size, it cannot disappear merely because the centered dictionary is enlarged. The correct inclusion is the central frequency inclusion, and it preserves the represented physical test exactly. Thus

- (2)
$$\begin{array}{c} \text{one bad centered size} \\ \Downarrow \\ \text{every larger centered size is bad at the same aperture.} \end{array}$$

The main contributions are the following.

- (i) A fixed-support quantitative estimate for the genuine complete zeta Weil form in which one factor is controlled by an L^1 norm plus a second-derivative L^1 norm.
- (ii) A summability-safe exact diagonal perturbation identity and a quantitative self-form error bound.
- (iii) A localized finite Fourier approximation theorem with simultaneous value, first-jet, and second-jet control on a fixed aperture.
- (iv) An explicit three-mode correction which enforces the centered moment conditions $M_0 = M_1 = M_2 = 0$ and yields legal globally C^2 hard-window tests while retaining the exact $L^1 + L^1$ second-derivative topology required in (i).
- (v) A strict finite sign-transfer theorem: every negative strict-aperture complete Weil test has a legal finite boundary-flat approximation with negative complete Weil value.
- (vi) The finite detection theorem (1) for the canonical direct-source finite matrix.
- (vii) Exact rank three of the boundary-moment map for $N \geq 1$, hence

$$\dim_{\mathbb{C}} V_N = 2N - 2$$

and a mode floor $N \geq 2$ for any nonzero constrained obstruction.

- (viii) Exact centered finite- N nesting preserving every centered moment, the represented finite function, and the canonical quadratic value, yielding the persistent negative tail (2).

Claim boundary. Nothing here proves the Riemann hypothesis. We do not prove that the canonical finite matrices are positive on the constrained sectors, do not extract a finite-to-infinite spectral limit, do not prove an eigenvalue-count theorem, and do not provide an effective threshold N_0 . The finite positivity criterion obtained in Section 9 is explicitly RH-equivalent: its value is structural compression of a hypothetical counterexample, not an unconditional proof of the positive direction. We also do not claim novelty for Fourier density, finite-dimensional rank-nullity, zero extension, or the abstract fact that a negative quadratic value remains negative when an embedding preserves that value. The novelty question concerns their exact realization and composition in the canonical Weil–CCM hierarchy.

2. CONVENTIONS AND THE CANONICAL FINITE HIERARCHY

Let $\mathcal{Z}$ denote the set of distinct nontrivial zeros of the Riemann zeta function, and let $m_\rho \geq 1$ be the multiplicity of $\rho \in \mathcal{Z}$. Write

$$(3) \quad \gamma_\rho := \frac{\rho - \frac{1}{2}}{i}.$$

We use the Fourier convention

$$(4) \quad \widehat{f}(z) := \int_{\mathbb{R}} f(x) e^{izx} dx.$$

For compactly supported C^2 tests, the complete Weil sesquilinear form is written in the zero-side convention

$$(5) \quad W(f, g) := \sum_{\rho \in \mathcal{Z}} m_\rho \widehat{f}(\gamma_\rho) \overline{\widehat{g}(\gamma_\rho)},$$

whenever the sum is defined. This is the convention used in the companion two-translate theorem core [8]. Reflection of the zero configuration gives Hermitian symmetry, and in particular $W(f, f) \in \mathbb{R}$.

Fix $L > 0$. For $n \in \mathbb{Z}$, define the normalized Fourier mode

$$(6) \quad U_{n,L}(x) := L^{-1/2} \exp\left(\frac{2\pi i n x}{L}\right).$$

For $N \in \mathbb{N}$ and coefficients

$$u = (u_{-N}, \dots, u_N) \in \mathbb{C}^{2N+1},$$

set

$$(7) \quad q_{N,L,u}(x) := \sum_{n=-N}^N u_n U_{n,L}(x)$$

and let

$$(8) \quad p_{N,L,u}(x) := \mathbf{1}_{[0,L]}(x) q_{N,L,u}(x)$$

be the hard-window zero extension.

Definition 2.1 (Centered moments and boundary-flat sector). *For $k \geq 0$ define*

$$(9) \quad M_k^{(N)}(u) := \sum_{n=-N}^N n^k u_n.$$

The boundary-flat coefficient space is

$$(10) \quad V_N := \{u \in \mathbb{C}^{2N+1} : M_0^{(N)}(u) = M_1^{(N)}(u) = M_2^{(N)}(u) = 0\}.$$

The terminology is literal. Since

$$q_{N,L,u}^{(j)}(0) = L^{-1/2} \left(\frac{2\pi i}{L} \right)^j M_j^{(N)}(u) \quad (j = 0, 1, 2),$$

and the finite Fourier formula is L -periodic together with its derivatives, $u \in V_N$ forces the value and first two jets to vanish at both endpoints. Therefore the hard-window vector (8) is globally C^2 .

The deterministic matrix used below is the *canonical direct-source* finite matrix, denoted

$$(11) \quad \mathcal{A}_{N,L} \in M_{2N+1}(\mathbb{C}).$$

It is the cutoff-free/direct equation-(4.4) normalization of the CCM finite source matrix. The distinction from the historical printed diagonal normalization and the exact scalar reconciliation between the two conventions are treated separately in [10]. The canonical object is also exactly the deterministic finite dictionary matrix.

For $u \in \mathbb{C}^{2N+1}$ write

$$(12) \quad Q_{N,L}^{\text{can}}(u) := u^* \mathcal{A}_{N,L} u.$$

Proposition 2.2 (Exact finite boundary-flat bridge). *Let $L > 0$, $N \in \mathbb{N}$, and $u \in V_N$. Then*

$$(13) \quad W(p_{N,L,u}, p_{N,L,u}) = Q_{N,L}^{\text{can}}(u).$$

Proof. The moment conditions make $p_{N,L,u}$ a compactly supported global C^2 test. The zero-side evenization theorem identifies its genuine complete zeta Weil self-form with the localized additive explicit-formula right-hand side. The finite dictionary identity identifies that additive right-hand side with the cutoff-free deterministic quadratic form, and the normalization reconciliation identifies the cutoff-free matrix with the canonical direct-source matrix. This is the formal composition

$$\text{zeta_W_boundaryFlatFiniteVector_eq_canonicalSourceQuadraticForm.}$$

□

3. QUANTITATIVE CONTINUITY OF THE COMPLETE WEIL FORM

The analytic input is deliberately asymmetric. Only one Fourier factor is integrated by parts twice; the other is estimated directly in L^1 . This produces one inverse-square zero weight and hence one fixed summable majorant.

Definition 3.1 (Inverse-square zeta-zero mass). *Set*

$$(14) \quad S_\zeta := \sum_{\rho \in \mathcal{Z}} \frac{m_\rho}{1 + |\gamma_\rho|^2}.$$

Standard zero counting implies $S_\zeta < \infty$; the corresponding summability statement is theorem-locked in the formal development.

Theorem 3.2 (Common-support Weil bound). *Let $\Lambda \geq 0$. Suppose $f \in C^2(\mathbb{R}; \mathbb{C})$ and $g \in C(\mathbb{R}; \mathbb{C})$ satisfy*

$$\text{supp } f, \text{supp } g \subseteq [-\Lambda, \Lambda].$$

Then

$$(15) \quad \|W(f, g)\| \leq e^\Lambda S_\zeta \left(\int_{\mathbb{R}} |f| + \int_{\mathbb{R}} |f''| \right) \left(\int_{\mathbb{R}} |g| \right).$$

Proof. For $z \in \mathbb{C}$, compact support gives the direct estimate

$$(16) \quad |\widehat{g}(z)| \leq e^{|\text{Im } z| \Lambda} \|g\|_1.$$

For the C^2 function f , the direct zeroth-order estimate and the twice-integrated-by-parts second-order estimate can be added before division, giving

$$(17) \quad |\widehat{f}(z)| \leq \frac{e^{|\text{Im } z| \Lambda}}{1 + |z|^2} (\|f\|_1 + \|f''\|_1).$$

For nontrivial zeta zeros, $|\text{Im } \gamma_\rho| \leq \frac{1}{2}$, and the same bound holds for $\overline{\gamma_\rho}$. Hence the two exponential factors contribute at most e^Λ . Every zero-side summand is therefore bounded by

$$e^\Lambda (\|f\|_1 + \|f''\|_1) \|g\|_1 \frac{m_\rho}{1 + |\gamma_\rho|^2}.$$

Summation and (14) give (15). $\square$

Proposition 3.3 (Exact diagonal cross identity). *Let $p, h \in C_c^2(\mathbb{R}; \mathbb{C})$. Then*

$$(18) \quad W(p, p) - W(h, h) = W(p - h, p) + W(h, p - h).$$

Proof. The identity is pointwise sesquilinear algebra on each zero summand. All four zero-side families are summable for compactly supported C^2 tests, so subtraction and addition may be transported through their sums. The summability step is part of the formal statement, rather than an implicit rearrangement of conditionally convergent series. $\square$

Corollary 3.4 (Self-form perturbation bound). *Assume $L > 0$, $p, h \in C_c^2(\mathbb{R}; \mathbb{C})$, and*

$$\text{supp } p, \text{supp } h \subseteq [-L, L].$$

Then

$$(19) \quad \|W(p, p) - W(h, h)\| \leq e^L S_\zeta \left[(\|p - h\|_1 + \|(p - h)''\|_1) \|p\|_1 + (\|h\|_1 + \|h''\|_1) \|p - h\|_1 \right].$$

In particular, if $0 < \eta \leq 1$ and

$$(20) \quad \|p - h\|_1 \leq \eta, \quad \|(p - h)''\|_1 \leq \eta,$$

then

$$(21) \quad \|W(p, p) - W(h, h)\| \leq e^L S_\zeta \eta (3\|h\|_1 + \|h''\|_1 + 2).$$

Proof. Apply Proposition 3.3 and Theorem 3.2 to the two cross terms. Under (20), the elementary bound

$$\|p\|_1 \leq \|h\|_1 + 1$$

gives (21) after collecting terms. $\square$

Remark 3.5. *The estimate is static and quantitative. No family dominated-convergence theorem is assumed. This distinction is useful below: once a finite approximation is made small in exactly the two error quantities in (20), strict negativity transfers immediately.*

4. LOCALIZED FOURIER APPROXIMATION WITH TWO-JET CONTROL

The first approximation step is formula-level and periodic. It does *not* yet claim that the hard-window zero extension is globally C^2 .

Lemma 4.1 (Strict support kills endpoint jets). *Let $L > 0$ and $h \in C^2(\mathbb{R}; \mathbb{C})$ satisfy*

$$\text{tsupp } h \subseteq (0, L).$$

Then

$$(22) \quad h(0) = h(L) = h'(0) = h'(L) = h''(0) = h''(L) = 0.$$

Proof. The closed support is contained in the open interval, so h vanishes on neighborhoods of both endpoints. The first two derivatives vanish there as well. $\square$

Identify the interval endpoints to obtain the circle $\mathbb{R}/L\mathbb{Z}$. By Lemma 4.1, the ordinary second derivative h'' descends continuously to that circle. Moreover,

$$(23) \quad \int_0^L h''(x) dx = h'(L) - h'(0) = 0,$$

so its zeroth Fourier coefficient vanishes.

Theorem 4.2 (Localized finite approximation with two-jet control). *Let $L > 0$, $h \in C^2(\mathbb{R}; \mathbb{C})$, and $\text{tsupp } h \subseteq (0, L)$. For every $\varepsilon > 0$ there exist $N \geq 1$ and $u \in \mathbb{C}^{2N+1}$ such that*

$$(24) \quad q_{N,L,u}(0) = 0$$

and, uniformly for $x \in [0, L]$,

$$(25) \quad |q_{N,L,u}(x) - h(x)| < \varepsilon,$$

$$(26) \quad |q'_{N,L,u}(x) - h'(x)| < \varepsilon,$$

$$(27) \quad |q''_{N,L,u}(x) - h''(x)| < \varepsilon.$$

Proof. Let H be the continuous periodic function induced by h'' . By (23), its zeroth Fourier coefficient is zero. Finite trigonometric polynomials are uniformly dense in the continuous functions on the circle, and deleting the constant coefficient from a sufficiently close approximant changes the error by at most the size of that coefficient. Thus one obtains a finite Fourier polynomial

$$P(x) = \sum_{0 < |n| \leq N} c_n e^{2\pi i n x / L}$$

which uniformly approximates H and has no zero mode.

For $n \neq 0$, divide the coefficient c_n by $(2\pi i n / L)^2$ and convert to the normalization (6). The resulting finite Fourier formula q_0 satisfies $q_0'' = P$. Changing only the zero-frequency coefficient anchors the function at the left endpoint without changing either derivative; call the anchored function q . Periodicity gives $q(L) = q(0) = 0$.

Set $e_1 = q' - h'$. Then $e'_1 = q'' - h''$ is uniformly small, while

$$\int_0^L e_1(x) dx = (q(L) - q(0)) - (h(L) - h(0)) = 0.$$

A vector-valued mean-zero derivative estimate gives

$$(28) \quad \sup_{[0,L]} |e_1| \leq 2L \sup_{[0,L]} |e'_1|.$$

Finally $e_0 = q - h$ satisfies $e_0(0) = 0$ and $e'_0 = e_1$, hence

$$(29) \quad \sup_{[0,L]} |e_0| \leq L \sup_{[0,L]} |e_1|.$$

Choosing the initial approximation tolerance small enough in terms of L and ε yields (25)–(27). $\square$

Remark 4.3. *The theorem is intentionally formula-level. The periodic finite formula may have small but nonzero endpoint first and second jets, so its hard zero extension need not yet be globally C^2 . The next section repairs exactly that seam.*

5. EXACT BOUNDARY-FLAT LEGALIZATION

For $N \geq 1$, the centered modes $-1, 0, 1$ are always available. They provide an explicit finite-rank correction for all three endpoint moments.

Definition 5.1 (Three-mode correction). *For $u \in \mathbb{C}^{2N+1}$ define $C_N u \in \mathbb{C}^{2N+1}$ to vanish outside the centered modes $-1, 0, 1$ and set*

$$(30) \quad (C_N u)_{-1} := \frac{M_1^{(N)}(u) - M_2^{(N)}(u)}{2},$$

$$(31) \quad (C_N u)_0 := M_2^{(N)}(u) - M_0^{(N)}(u),$$

$$(32) \quad (C_N u)_1 := -\frac{M_1^{(N)}(u) + M_2^{(N)}(u)}{2}.$$

Define the boundary-flat projection

$$(33) \quad P_N u := u + C_N u.$$

Lemma 5.2 (Exact moment cancellation). *For every $N \geq 1$ and $u \in \mathbb{C}^{2N+1}$,*

$$(34) \quad M_k^{(N)}(P_N u) = 0, \quad k = 0, 1, 2.$$

Thus $P_N u \in V_N$.

Proof. A direct substitution of (30)–(32) gives

$$M_k^{(N)}(C_N u) = -M_k^{(N)}(u), \quad k = 0, 1, 2.$$

Additivity of the centered moments gives the result. $\square$

The correction is not merely exact; it becomes quantitatively small when the three endpoint moments are small. Write

$$(35) \quad \omega_L := \frac{2\pi i}{L}.$$

For the formula $q = q_{N,L,u}$,

$$(36) \quad \begin{aligned} q(0) &= L^{-1/2} M_0^{(N)}(u), \\ q'(0) &= L^{-1/2} \omega_L M_1^{(N)}(u), \\ q''(0) &= L^{-1/2} \omega_L^2 M_2^{(N)}(u). \end{aligned}$$

Hence anchoring already gives $M_0 = 0$, and two-jet approximation to a strict-collar target makes M_1 and M_2 arbitrarily small.

Lemma 5.3 (Correction bounds). *Let $L > 0$, $N \geq 1$, and suppose $M_0^{(N)}(u) = 0$. Then for every $x \in \mathbb{R}$,*

$$(37) \quad |q_{N,L,C_N u}(x)| \leq L^{-1/2} (|M_1^{(N)}(u)| + 2|M_2^{(N)}(u)|),$$

$$(38) \quad |q''_{N,L,C_N u}(x)| \leq L^{-1/2} |\omega_L|^2 (|M_1^{(N)}(u)| + |M_2^{(N)}(u)|).$$

Proof. Use (30)–(32), $M_0 = 0$, and the constant modulus $|U_{n,L}(x)| = L^{-1/2}$ for the three reserved modes. After two derivatives the zero mode disappears and the two nonzero reserved modes both acquire the factor $|\omega_L|^2$. $\square$

Theorem 5.4 (Boundary-flat approximation in the Weil topology). *Let $L > 0$, $h \in C^2(\mathbb{R}; \mathbb{C})$, and $\text{tsupp } h \subseteq (0, L)$. For every $\eta > 0$ there exist $N \geq 1$ and $u \in V_N$ such that $p := p_{N,L,u}$ is a compactly supported global C^2 test and*

$$(39) \quad \|p - h\|_1 < \eta, \quad \|(p - h)''\|_1 < \eta.$$

Proof. Apply Theorem 4.2 with a uniform tolerance ε to be chosen. Let v be the resulting anchored coefficient vector. By (24) and (36), $M_0^{(N)}(v) = 0$. Lemma 4.1 and the uniform first- and second-jet estimates imply

$$|M_1^{(N)}(v)| \leq \frac{\sqrt{L}}{|\omega_L|} \varepsilon, \quad |M_2^{(N)}(v)| \leq \frac{\sqrt{L}}{|\omega_L|^2} \varepsilon.$$

Set $u = P_N v$. Lemma 5.2 gives $u \in V_N$, so the hard-window vector $p_{N,L,u}$ is globally C^2 .

On $[0, L]$, the correction is controlled by Lemma 5.3; outside $[0, L]$, both p and h vanish. Therefore there are explicit constants $A_0(L), A_2(L) > 0$ such that

$$\sup_{\mathbb{R}} |p - h| \leq A_0(L) \varepsilon, \quad \sup_{\mathbb{R}} |(p - h)''| \leq A_2(L) \varepsilon.$$

The common support has length L , so

$$\|p - h\|_1 \leq L A_0(L) \varepsilon, \quad \|(p - h)''\|_1 \leq L A_2(L) \varepsilon.$$

Choosing ε smaller than $\eta/[L \max(A_0(L), A_2(L))]$ gives (39). The formal proof retains explicit constants throughout; no compactness or nonconstructive density argument is used at this legalization stage. $\square$

6. STRICT FINITE SIGN TRANSFER

We now connect a hypothetical off-critical zero to the finite hierarchy. The only upstream input concerning the zero is a negative complete Weil test.

Proposition 6.1 (Strict-aperture negative complete test). *Let $\rho_0 \in \mathcal{Z}$ satisfy*

$$(40) \quad \operatorname{Re} \rho_0 \neq \frac{1}{2}.$$

Then there exist $L > 0$ and $h \in C_c^2(\mathbb{R}; \mathbb{C})$ such that

$$(41) \quad \operatorname{tsupp} h \subseteq (0, L)$$

and

$$(42) \quad \operatorname{Re} W(h, h) < 0.$$

Proof. The complete two-translate theorem gives a compactly supported C^2 test f , a translate $T_t f$, and a vector $(a, b) \in \mathbb{C}^2$ for which the complete two-translate matrix has negative quadratic value [8]. Because W is linear in its first argument and conjugate-linear in its second, the physical combination

$$h_0 = \bar{a} f + \bar{b} T_t f$$

contracts that matrix exactly:

$$W(h_0, h_0) = (a, b)^* G_f(t) (a, b) < 0.$$

It remains compactly supported and C^2 . Choose $r > 0$ with $\operatorname{tsupp} h_0 \subseteq (-r, r)$ and translate right by $2r$. The new support lies in $(r, 3r) \subset (0, 4r)$. Common translation invariance of the complete Weil form preserves the negative self-value. Set $L = 4r$. $\square$

Theorem 6.2 (Strict finite sign transfer). *Let $L > 0$ and let $h \in C_c^2(\mathbb{R}; \mathbb{C})$ satisfy $\operatorname{tsupp} h \subseteq (0, L)$ and $\operatorname{Re} W(h, h) < 0$. Then there exist $N \geq 1$ and $u \in V_N$ such that*

$$(43) \quad \operatorname{Re} W(p_{N,L,u}, p_{N,L,u}) < 0.$$

Proof. Set

$$m := -\operatorname{Re} W(h, h) > 0, \quad H_0 := \|h\|_1, \quad H_2 := \|h''\|_1,$$

and

$$(44) \quad K := e^L S_\zeta (3H_0 + H_2 + 2).$$

Choose $0 < \eta \leq 1$ so that $K\eta < m$; for example

$$\eta = \min \left(1, \frac{m}{2(K+1)} \right).$$

By Theorem 5.4, choose a legal finite $p = p_{N,L,u}$ with $u \in V_N$ and the two errors in (39) smaller than η . Corollary 3.4 gives

$$\|W(p, p) - W(h, h)\| \leq K\eta < m.$$

Hence

$$\operatorname{Re} W(p, p) \leq \operatorname{Re} W(h, h) + |W(p, p) - W(h, h)| < -m + m = 0.$$

$\square$

Theorem 6.3 (Finite canonical detection). *If $\rho_0 \in \mathcal{Z}$ satisfies $\operatorname{Re} \rho_0 \neq \frac{1}{2}$, then there exist $L > 0$, $N \geq 1$, and $u \in V_N$ such that*

$$(45) \quad \operatorname{Re} Q_{N,L}^{\text{can}}(u) < 0.$$

In particular $u \neq 0$.

Proof. Proposition 6.1 gives a strict-aperture negative test. Theorem 6.2 transfers strict negativity to a boundary-flat finite vector. Proposition 2.2 identifies its complete Weil value with the canonical finite quadratic form. A zero vector has quadratic value zero, so the witness is nonzero. $\square$

7. EXACT DIMENSION OF THE BOUNDARY-FLAT SECTOR

The three moment constraints are genuinely independent as soon as the modes $-1, 0, 1$ are present.

Definition 7.1 (Boundary moment map). *For $N \geq 1$, define*

$$(46) \quad \mathcal{M}_N : \mathbb{C}^{2N+1} \longrightarrow \mathbb{C}^3, \quad \mathcal{M}_N(u) := (M_0^{(N)}(u), M_1^{(N)}(u), M_2^{(N)}(u)).$$

Proposition 7.2 (Exact rank three). *For every $N \geq 1$, the map $\mathcal{M}_N$ is surjective.*

Proof. Given $(m_0, m_1, m_2) \in \mathbb{C}^3$, use only the modes $-1, 0, 1$ and set

$$(47) \quad a = \frac{m_2 - m_1}{2}, \quad b = m_0 - m_2, \quad c = \frac{m_2 + m_1}{2}.$$

The coefficient vector supported by

$$u_{-1} = a, \quad u_0 = b, \quad u_1 = c$$

has moments m_0, m_1, m_2 . This is the inverse of the 3×3 Vandermonde system at $-1, 0, 1$. $\square$

Theorem 7.3 (Exact constrained dimension). *For $N \geq 1$,*

$$(48) \quad \dim_{\mathbb{C}} V_N = 2N - 2.$$

Consequently $V_1 = \{0\}$, and every nonzero vector in V_N with $N \geq 1$ satisfies $N \geq 2$.

Proof. By definition $V_N = \ker \mathcal{M}_N$. Proposition 7.2 gives $\text{rank } \mathcal{M}_N = 3$. Rank-nullity yields

$$\dim V_N = (2N + 1) - 3 = 2N - 2.$$

The remaining statements follow immediately. $\square$

Corollary 7.4 (Mode floor for finite detection). *The witness in Theorem 6.3 can always be chosen with*

$$(49) \quad N \geq 2.$$

8. EXACT CENTERED FINITE- N NESTING

The finite hierarchy is nested by centered frequency, not by prefix inclusion of coordinate labels. In the integer-index notation used here, the correct map is simply zero extension.

Definition 8.1 (Centered zero extension). *Let $N \leq M$. Define*

$$(50) \quad E_{N,M} : \mathbb{C}^{2N+1} \longrightarrow \mathbb{C}^{2M+1}$$

by

$$(51) \quad (E_{N,M}u)_n := \begin{cases} u_n, & |n| \leq N, \\ 0, & N < |n| \leq M. \end{cases}$$

Proposition 8.2 (All centered moments are preserved). *For every $k \geq 0$,*

$$(52) \quad M_k^{(M)}(E_{N,M}u) = M_k^{(N)}(u).$$

In particular,

$$(53) \quad u \in V_N \implies E_{N,M}u \in V_M.$$

Proof. All newly introduced coordinates vanish, while the shared coordinates carry the same centered integer labels. $\square$

Proposition 8.3 (Exact physical-function preservation). *For $N \leq M$,*

$$(54) \quad p_{M,L,E_{N,M}u} = p_{N,L,u}$$

pointwise on $\mathbb{R}$.

Proof. The larger sum contains exactly the same nonzero Fourier coefficients at the same integer frequencies, and all added coefficients are zero. The hard window is the same interval $[0, L]$. $\square$

The same centered identification occurs directly at matrix level.

Proposition 8.4 (Canonical principal-block nesting). *For $N \leq M$, the central $(2N+1) \times (2N+1)$ principal block of $\mathcal{A}_{M,L}$ is exactly $\mathcal{A}_{N,L}$.*

Proof. The canonical entry is a deterministic function of the two centered integer frequencies and L . The central embedding preserves both integer labels exactly, so every shared entry is unchanged. $\square$

Theorem 8.5 (Exact quadratic preservation). *Let $L > 0$ and $N \leq M$. Then for every $u \in \mathbb{C}^{2N+1}$,*

$$(55) \quad Q_{M,L}^{\text{can}}(E_{N,M}u) = Q_{N,L}^{\text{can}}(u).$$

Moreover, $E_{N,M}$ is injective and preserves the ordinary Euclidean norm of coefficient vectors.

Proof. Equation (55) follows immediately from Proposition 8.4 because all coefficients outside the central block vanish. Equivalently, one may combine Proposition 8.3 with the exact localized finite right-hand-side identity. Injectivity and Euclidean norm preservation are immediate from zero extension. $\square$

Remark 8.6 (Exact containment, not asymptotic approximation). *The larger centered problem does not merely approximate the old witness. It contains the identical physical hard-window test and assigns it the identical canonical quadratic value. This exactness is what makes the persistence theorem below stronger than a convergence statement.*

9. PERSISTENT FINITE DETECTION AND A FINITE POSITIVITY CRITERION

We can now compose the finite detection theorem with exact nesting.

Theorem 9.1 (Persistent finite detection). *Let $\rho_0 \in \mathcal{Z}$ satisfy*

$$\text{Re } \rho_0 \neq \frac{1}{2}.$$

Then there exist $L > 0$, an integer $N_0 \geq 2$, and a nonzero vector $u_0 \in V_{N_0}$ such that

$$(56) \quad \text{Re } Q_{N_0,L}^{\text{can}}(u_0) < 0.$$

For every $M \geq N_0$, the centered extension

$$(57) \quad u_M := E_{N_0, M} u_0$$

is nonzero, belongs to V_M , represents the same physical test, and satisfies the exact identity

$$(58) \quad Q_{M, L}^{\text{can}}(u_M) = Q_{N_0, L}^{\text{can}}(u_0).$$

Consequently,

$$(59) \quad \text{Re } Q_{M, L}^{\text{can}}(u_M) < 0 \quad (M \geq N_0).$$

Proof. Theorem 6.3 and Corollary 7.4 provide L, N_0, u_0 satisfying (56). Proposition 8.2 gives $u_M \in V_M$; injectivity of zero extension gives $u_M \neq 0$; Proposition 8.3 gives exact physical-test equality; and Theorem 8.5 gives (58). Taking real parts yields (59). $\square$

Corollary 9.2 (Upward-closed bad-size set). *For fixed $L > 0$, define*

$$(60) \quad \mathcal{B}_L := \{N \geq 1 : \exists 0 \neq u \in V_N, \text{Re } Q_{N, L}^{\text{can}}(u) < 0\}.$$

Then

$$(61) \quad N \in \mathcal{B}_L, N \leq M \implies M \in \mathcal{B}_L.$$

If an off-critical zero exists, $\mathcal{B}_L$ is nonempty for at least one aperture L and contains an infinite tail.

The same theorem yields a constrained finite positivity criterion.

Theorem 9.3 (Boundary-flat finite hierarchy criterion). *The following statements are equivalent.*

- (i) *Every nontrivial zero of ζ lies on the critical line.*
- (ii) *For every $L > 0$, every $N \geq 1$, and every $u \in V_N$,*

$$(62) \quad \text{Re } Q_{N, L}^{\text{can}}(u) \geq 0.$$

- (iii) *The same condition holds for every $L > 0$, every $N \geq 2$, and every $u \in V_N$.*

Proof. Assume (i). Then every γ_ρ is real. For any compactly supported C^2 test p ,

$$W(p, p) = \sum_{\rho \in \mathcal{Z}} m_\rho |\hat{p}(\gamma_\rho)|^2 \geq 0.$$

For $u \in V_N$, Proposition 2.2 gives (62). Thus (i) $\Rightarrow$ (ii), and (ii) $\Rightarrow$ (iii) is immediate.

If (iii) holds and an off-critical zero exists, Theorem 9.1 produces $L > 0$, $N_0 \geq 2$, and $u_0 \in V_{N_0}$ with negative real quadratic value, contradicting (iii). Hence (iii) $\Rightarrow$ (i). $\square$

Remark 9.4 (Logical strength). *Theorem 9.3 is RH-equivalent. It does not supply the universal nonnegativity in (ii) or (iii). Its content is that a hypothetical counterexample cannot remain hidden outside the legal centered finite hierarchy: it forces a negative finite witness, and Theorem 9.1 makes that witness persistent at one fixed aperture.*

10. RELATION TO EXISTING FINITE WEIL CONSTRUCTIONS

The broad implication “off-critical behavior creates negative Weil geometry” is classical. The point of the present theorem is the exact finite object in which the negativity is forced and the exact persistence law once it appears.

10.1. Bombieri’s finite negative-inertia theory. Bombieri studies finite systems built from transformed zero configurations and proves precise negative-inertia statements for complex-conjugate pairs [1]. Under additional hypotheses, including finiteness conditions on the off-critical population in the zeta passage, this yields eventual finite negative-inertia information. The present centered hierarchy is indexed instead by deterministic Fourier modes $-N, \dots, N$. No identification of Bombieri’s finite matrices with $\mathcal{A}_{N,L}$ is used.

10.2. Finite Guinand–Weil dictionaries. Groskin constructs an exact finite transport from real-even centered Galerkin coefficient vectors to band-limited Guinand–Weil tests and proves that every corresponding real-even finite quadratic value equals an exact complete zero sum [6]. That result supplies an exact finite measurement device. The present paper addresses a different direction: starting from one arbitrary hypothetical off-critical zeta zero, it proves that some legal canonical centered finite vector must acquire a negative value and that the same negative vector embeds exactly into every larger centered truncation at the same aperture.

10.3. Connes–van Suijlekom, CCM, and operator formulations. Connes and van Suijlekom derive structured finite matrices from lower-bounded quadratic forms and use finite-to-continuous zero-localization arguments [5]. Connes, Consani, and Moscovici use centered low-frequency spaces in their zeta spectral-triple program [4]. Suzuki develops a localized operator realization of Weil’s quadratic form through the screw function [12]. These works supply the natural finite and operator context for the canonical matrix but do not replace the approximation-and-persistence theorem proved here.

10.4. Comparison table. Table 1 summarizes the closest finite comparison. The entries are deliberately narrow; “not this formulation” does not mean that the cited work lacks related spectral information.

TABLE 1. Comparison with the closest finite frameworks.

Feature	Bombieri 2000	Groskin 2026	Present result
Finite index	transformed zero configuration	real-even centered Fourier/Galerkin band	centered Fourier band
Exact finite zero-sum dictionary on the corresponding Galerkin sector	different finite object	yes (real-even)	inherited/used
One arbitrary off-critical zeta zero forces a negative present canonical finite vector	not this formulation	not the stated endpoint	yes
Three exact boundary moments in the negative witness	not this formulation	not the stated endpoint	yes
Fixed aperture after first detection	not this formulation	not applicable to a counterexample theorem	yes
Same witness embeds to every larger centered size	not this formulation	not the stated endpoint	yes
Quadratic value unchanged under centered extension	not this formulation	not the stated endpoint	yes
Lean formalization of detection/persistence core	no	no	yes

11. FORMAL VERIFICATION IN LEAN 4

The detection-and-persistence theorem core is formalized in the public repository

<https://github.com/HMarcusWH/zeta-23-lean>.

The paper is pinned to main commit

`c7129b1856ea03cdf8b831ae1424140f8a7d90a9`,

with Git tree

`56d082947fac6eb4666d0e0666e2c8bcd3c0a7e8`.

The toolchain is Lean `v4.33.0-rc2`; Mathlib is pinned to

`51e6992efd06126df61a496bebf8f49482a4e129`.

The main paper-to-Lean correspondence is recorded in Table 2. Some paper statements are direct compositions of more primitive declarations rather than separately named wrappers; this is stated explicitly rather than treating a paper corollary as a distinct compiler object.

Formal-status boundary. Theorem 9.1 in the raw coefficient language is a direct paper composition of the formally verified finite-detection, exact-dimension, and centered-nesting declarations listed below; the repository also contains the Euclidean fixed-aperture persistence wrapper. Theorem 9.3 is not claimed to have a separate Lean declaration at this snapshot. Its off-critical contradiction direction is an immediate consequence of the formalized detection theorem, while its critical-line direction is the elementary zero-side modulus-square identity displayed in its proof. Thus the machine-checked claim is the detection-and-persistence core, not a separately named formalization of the three-way RH-equivalent criterion.

TABLE 2. Paper results and Lean declarations.

Paper result	Lean declaration
Inverse-square zero-mass summability	<code>zeta_invSqZeroWeight_summable</code>
Common-support Weil estimate	<code>zeta_W_norm_le_commonSupport</code>
Exact diagonal cross identity	<code>zeta_W_self_sub_self_eq_cross</code>
Self-form common-support perturbation bound	<code>zeta_W_self_sub_self_norm_le_commonSupport</code>
Localized value/first/second-jet approximation	<code>exists_localizedFinite_uniform_C2_approx</code>
Exact boundary-flat finite approximation in the consumed topology	<code>exists_boundaryFlatFinite_WCONT_approx</code>
Global C^2 legality of boundary-flat hard-window vectors	<code>contDiff_localizedFiniteVector_of_boundaryFlat</code>
Genuine Weil form equals canonical finite quadratic form	<code>zeta_W_boundaryFlatFiniteVector_eq_canonicalSourceQuadraticForm</code>
Strict finite sign transfer	<code>exists_boundaryFlatFinite_negativeW_of_strictAperture</code>
Off-line zero $\Rightarrow$ canonical finite negative obstruction	<code>exists_boundaryFlat_negativeCanonicalSourceQuadraticForm_of_offLine_zero</code>
Boundary moment map surjectivity	<code>boundaryMomentTripleMap_surjective</code>
Exact dimension $\dim V_N = 2N - 2$	<code>finrank_boundaryFlatSubspace</code>
Nonzero constrained vector $\Rightarrow N \geq 2$	<code>two_le_of_ne_zero_mem_boundaryFlatSubspace</code>
Centered integer label preserved by embedding	<code>centeredIndex_centeredEmbedding</code>
Canonical central principal-block identity	<code>canonicalSourceMatrix_submatrix_centeredEmbedding</code>
Every centered moment preserved	<code>centeredMoment_centeredZeroExtend</code>
Represented hard-window finite function preserved	<code>localizedFiniteVector_centeredZeroExtend</code>
Canonical quadratic value preserved	<code>quadraticForm_canonicalSourceMatrix_centeredZeroExtend</code>
Persistent negative Euclidean constrained obstruction	<code>negativeEuclideanConstrained_persists_of_le</code>
Fixed-aperture negative tail from an off-line zero	<code>exists_fixedAperture_forall_ge_ne_zero_mem_euclideanBoundaryFlatSubspace_negativeCanonicalInnerSelf_of_offLine_zero</code>

The relevant source blobs at the pinned snapshot include

<code>Zeta23/ExceptionalZero/WeilContinuity.lean</code>	893175d5d5777e9737bfac7a5af8e2c5bd8a2612
<code>Zeta23/CCM/LocalizedFourierApproximation.lean</code>	e5bd6c64fdb622796a39714c31a728956b1eb10a
<code>Zeta23/CCM/BoundaryFlatApproximation.lean</code>	affdfea9181f7a9310c3b1fced16d14c1573c797
<code>Zeta23/ExceptionalZero/FiniteNegativeObstruction.lean</code>	acba8c5cba15a59c7d3c66cd966a551421150b49
<code>Zeta23/CCM/ConstrainedEuclideanSector.lean</code>	2ef60150a02c615d1c5179d8880d8c402749d012
<code>Zeta23/CCM/NestedFinite.lean</code>	804816b8650482ecd65b48d6d822d16e89c88ef7
<code>Zeta23/ExceptionalZero/NestedEuclideanObstruction.lean</code>	d0073491e08257130f638cf6189b7c7877688c9e

The strict-aperture negative-test transport is in `Zeta23/ExceptionalZero/NegativeWeilTestSupport.lean` (blob `c544c13b92e31463b880bd93ca1885ca41122de0`), and the exact boundary-flat finite bridge is in `Zeta23/ExceptionalZero/BoundaryFlatFiniteWeil.lean` (blob `d456cf86c558c76c87867fa78c98538f8d1617c5`).

The repository workflow builds the promoted CCM and exceptional-zero umbrellas and rejects occurrences of `sorry` and explicit project `axiom` declarations in those promoted subtrees. The substantive theorem files also contain explicit `#print axioms` checks. The promoted endpoints used here report only the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`. These logical dependencies are not mathematical assumptions about zeta zeros or RH.

12. PRIORITY BOUNDARY AND NOVELTY STATEMENT

The broad background is classical. Weil positivity is equivalent to RH; hypothetical off-critical spectral data is associated with negative Weil geometry; smooth compactly supported functions admit finite Fourier approximation; three independent linear constraints cut finite dimension by three; and exact zero extension trivially preserves an already-existing coefficient vector. Bombieri’s work already establishes finite negative-index phenomena in a different finite spectral model [1], while Groskin establishes an exact finite Guinand–Weil dictionary for the real-even centered finite setting [6]. No novelty claim is made for any of those facts in isolation.

The narrower claim concerns the following composition in the canonical centered Weil–CCM hierarchy:

$$\begin{array}{c}
 \text{one arbitrary hypothetical off-critical zeta zero} \\
 \Downarrow \\
 \text{one strict negative complete Weil test} \\
 \Downarrow \\
 (63) \quad \text{one legal three-moment finite canonical negative vector} \\
 \Downarrow \\
 \text{the same physical vector at every larger centered size} \\
 \text{with unchanged canonical quadratic value.}
 \end{array}$$

In a targeted source-level audit of the closest materials available to the author—including Bombieri’s finite-inertia theory, the Connes–van Suijlekom/CCM finite constructions, Groskin’s exact dictionary, Suzuki’s operator formulation, and the companion finite manuscripts—we did not identify a prior theorem combining all three arrows in (63). We therefore describe the result conservatively as an *apparently new finite counterexample-detection and persistence theorem* for the canonical centered hierarchy. This is a “to our knowledge” statement, not a claim of exhaustive universal priority.

There is also an important boundary relative to the earlier scalar-defect manuscript. Its open problem P4 asked for quantitative finite detection, including control of how large the finite dictionary must be [9]. The present paper closes the *existence* part:

$$\exists N_0.$$

It does not provide an effective formula or usable numerical upper bound for N_0 . That effective threshold remains open.

13. OPEN PROBLEMS

The persistence theorem removes finite-size selection as a hiding mechanism once a counterexample has entered the hierarchy. Several sharper questions remain.

P1. Effective finite threshold. Given the strict negative margin

$$m = -\operatorname{Re} W(h, h) > 0,$$

the aperture L , and quantitative information on the detector, derive an explicit upper bound

$$N_0 \leq F(L, m, \text{detector data})$$

for a finite negative witness. The proof above is existential because the Fourier-density step is not packaged as an effective mode-count estimate.

P2. First bad constrained size and scalar instability. For a fixed aperture with $\mathcal{B}_L \neq \emptyset$, let $N_*(L)$ be its least bad size. The current formal development now also contains exact reversal symmetry, equal constrained parity dimensions, and one-dimensional parity successor shells. The next non-repackaging problem is to combine those known facts with first badness and derive an exact one-dimensional Schur, Feshbach, KKT, or secular condition for the first finite instability. A one-dimensional shell alone does not imply that the shell vector itself is negative because coupling to the old nonnegative sector may remain.

P3. Constrained spectral extraction. Package the canonical matrix as the self-adjoint operator compressed to V_N , derive a negative constrained eigenmode from a negative Rayleigh direction, and compare the resulting KKT equations with the exact low-rank displacement and moment-flag identities already available in the repository. This is finite spectral structure; it must not be conflated with a finite-to-infinite RH argument.

P4. General L -functions. Determine which other explicit-formula settings admit the same four ingredients: a complete negative detector from off-line spectral data, a legal finite core defined by finitely many boundary moments, an exact finite dictionary, and centered nesting that preserves the physical test and quadratic value.

P5. Minimal analytic topology. The proof uses the asymmetric topology

$$\|p - h\|_1 + \|(p - h)''\|_1.$$

Characterize the weakest natural form-domain topology in which complete Weil negativity transfers to this finite dictionary, and determine whether the constants in Theorem 3.2 can be sharpened without assuming information equivalent to RH.

14. CONCLUSION

An exact finite dictionary is a representation theorem: it says what a finite quadratic value means on the zero side. It does not by itself say that a hypothetical counterexample to the Riemann hypothesis must be visible in any finite member of that dictionary. The present paper supplies that missing direction.

A single hypothetical off-critical zero first produces a negative complete Weil test. A fixed-support quantitative estimate shows that strict negativity is stable under approximation in an L^1 plus second-derivative L^1 topology. Localized Fourier approximation provides the required finite formulas, and an explicit correction in the three modes $-1, 0, 1$ enforces the exact boundary moments needed for a legal global C^2 hard-window test. The exact finite dictionary then converts the surviving negative Weil value into a negative canonical finite matrix direction.

The finite obstruction is not isolated. The boundary-flat sector has dimension $2N - 2$, and centered zero extension preserves every moment, the physical test, and the canonical quadratic value exactly. Thus one finite negative witness at one aperture persists unchanged through every larger centered truncation. Equivalently, universal nonnegativity of the canonical quadratic forms on all boundary-flat finite sectors is a finite hierarchy criterion for the critical-line statement.

None of this establishes that universal nonnegativity. The Riemann hypothesis remains open. The result instead localizes exactly where a hypothetical failure must appear and proves that, once it appears in the canonical centered hierarchy, increasing the finite dimension cannot make it disappear.

APPENDIX A. THREE-MODE CORRECTION ALGEBRA

For completeness, write

$$m_k := M_k^{(N)}(u), \quad k = 0, 1, 2.$$

The correction from Definition 5.1 has coefficients

$$a = \frac{m_1 - m_2}{2}, \quad b = m_2 - m_0, \quad c = -\frac{m_1 + m_2}{2}$$

at the frequencies $-1, 0, 1$. Its first three moments are

$$M_0(C_N u) = a + b + c = -m_0,$$

$$M_1(C_N u) = -a + c = -m_1,$$

$$M_2(C_N u) = a + c = -m_2.$$

Thus $M_k(P_N u) = 0$ for $k = 0, 1, 2$.

If $m_0 = 0$, then

$$|a| + |b| + |c| \leq |m_1| + 2|m_2|,$$

which gives (37). After two derivatives the zero-frequency term disappears, while both ± 1 modes gain the factor ω_L^2 , and

$$|a| + |c| \leq |m_1| + |m_2|,$$

giving (38). These elementary inequalities are the quantitative reason that enforcing exact boundary legality does not destroy a sufficiently accurate finite approximation.

APPENDIX B. REPRODUCIBILITY AND DISCLOSURES

Repository. <https://github.com/HMarcusWH/zeta-23-lean>

Proof-bearing source snapshot. `c7129b1856ea03cdf8b831ae1424140f8a7d90a9`

Tree. `56d082947fac6eb4666d0e0666e2c8bcd3c0a7e8`

Lean. `leanprover/lean4:v4.33.0-rc2`

Mathlib. `51e6992efd06126df61a496bebf8f49482a4e129`

Data and code availability. No empirical dataset is required for the theorem. The Lean 4 formalization, finite normalization audits, theorem registries, and validation scripts are publicly available in the repository at the pinned source snapshot.

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Use of AI tools. Language-model tools were used for drafting, code assistance, proof organization, and literature-search support. The author takes responsibility for the mathematical statements, proofs, references, claim boundaries, and released formal verification. AI systems are not authors.

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